Regression Analysis of UCI Wine Quality (Red + White)

Design and Creative Technologies, Torrens University

- **Student:** Luis Guilherme de Barros Andrade Faria - A00187785
- **Subject Name:** Machine Learning
- **Subject Code:** MLN 601
- **Title:** Regression Analysis of UCI Wine Quality - multi-model comparison
- **Lecturer:** Dr. Kamran Shaukat
- **Assessment No.:** 1
- **Date:** June 2026

Extra details:

Field	Value
Dataset	UCI Wine Quality - red + white combined (Cortez et al., 2009)
Algorithms	Linear Regression (baseline) + Ridge, Lasso, ElasticNet, Random Forest
Tuning	GridSearchCV (5-fold cross-validation)
Methodology	CRISP-DM

This notebook follows the six CRISP-DM stages to predict the sensory **quality** score of wine from physicochemical measurements. It extends the linear-regression baseline by **comparing multiple regression models with tuned hyperparameters** and by using **both the red and white datasets combined**, so a larger, richer sample is available.

> Quality is a subjective, ordinal score, so every model is framed as an **interpretable decision-support tool**, not a replacement for expert tasting.

---

1. Business Understanding

1.1 Project objective

Wineries record objective chemical properties (acidity, sulphates, residual sugar, alcohol, etc.) during production. This project tests whether those measurements can predict the sensory **quality** score (0-10 scale; 3-9 observed across this sample). Both the **red and white** UCI datasets are combined into one table, with a `wine_type` indicator, so the model learns from a larger sample and can be compared against single-type models.

The task is **supervised regression**: 11 physicochemical attributes plus a `wine_type` flag as inputs, and `quality` as the dependent variable. A useful model would help a quality-control team screen batches for likely low/high quality before final expert tasting. Predictions are decision support, because quality is subjective and influenced by factors not present in the data.

Currently, wine quality is assessed by expert tasting, which is slow, costly, and subjective. This project follows the CRISP-DM methodology (Chapman et al., 2000) end to end, from business understanding through a reflective deployment stage.

1.2 Evaluation criteria

Criterion	Decision
Prediction error	MAE and RMSE, in quality-score units (lower is better)
Model fit	R^2 (variance explained on the held-out test set)
Baseline	Predict the training-set mean quality for every test row
Model selection	Train several regression models, tune their hyperparameters with cross-validation, and keep the lowest-RMSE model
Practical success	Beat the mean baseline and produce an interpretable, defensible best model
Communication	Explain results in plain language for a quality-control stakeholder

A perfect model is unrealistic, as **quality** is judged by humans and the dataset contains physicochemical variables only.

2. Data Understanding

2.1 Data acquisition

Item	Detail
Source	UCI Machine Learning Repository (id 186)
Dataset page	https://archive.ics.uci.edu/dataset/186/wine+quality
Selected files	`winequality-red.csv` and `winequality-white.csv`
Acquisition	Read the local CSVs downloaded from UCI (URL fallback kept for reproducibility)
Separator	`;`
Combination	Concatenate red + white, adding `wine_type` (1 = red, 0 = white)
Expected shape	~6,497 rows x 13 columns

The data was obtained from the UCI Machine Learning Repository (University of California, Irvine, n.d.); both the red and white variants (Cortez et al., 2009) are used. Dr. Kamran Shaukat confirmed that combining the two datasets gives a larger sample and is acceptable for this assessment, so both are merged with a wine-type indicator that lets the model account for systematic differences between the two.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.ensemble import RandomForestRegressor
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_scor

sns.set_theme(style="whitegrid")
RANDOM_STATE = 42
```

In [2]:
```
from pathlib import Path

# Resolve project folders relative to the notebook location so the same
# notebook runs locally and on Colab/Azure without editing paths.
NB_DIR = Path.cwd()
BASE_DIR = NB_DIR.parent if NB_DIR.name == "notebook" else NB_DIR
OUTPUT_DIR = BASE_DIR / "outputs"
FIG_DIR = OUTPUT_DIR / "figures"
FIG_DIR.mkdir(parents=True, exist_ok=True)

UCI_BASE = ("https://archive.ics.uci.edu/ml/machine-learning-databases/"
            "wine-quality/")


def load_wine(colour):
    fname = f"winequality-{colour}.csv"
    candidates = [BASE_DIR / "dataset" / fname,
                  NB_DIR / "dataset" / fname,
                  NB_DIR / fname]
    local = next((p for p in candidates if p.exists()), None)
    if local is not None:
        print("Loading local:", local)
        return pd.read_csv(local, sep=";")
    print("Loading from UCI URL:", UCI_BASE + fname)
    return pd.read_csv(UCI_BASE + fname, sep=";")


red = load_wine("red")
white = load_wine("white")
red["wine_type"] = 1    # 1 = red
white["wine_type"] = 0  # 0 = white
df = pd.concat([red, white], ignore_index=True)
```

```
print("Red:", red.shape, "| White:", white.shape, "| Combined:", df.shape)
df.head()
```

```
Loading local: /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/
assignments/Assessment1/dataset/winequality-red.csv
Loading local: /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/
assignments/Assessment1/dataset/winequality-white.csv
Red: (1599, 13) | White: (4898, 13) | Combined: (6497, 13)
```

Out[2]:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphat
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.

2.2 Variable description

Variable	Role	Type	Meaning / unit
fixed acidity	Feature	Continuous	Non-volatile acids (g/dm3)
volatile acidity	Feature	Continuous	Acetic acid (g/dm3)
citric acid	Feature	Continuous	Citric acid (g/dm3)
residual sugar	Feature	Continuous	Sugar after fermentation (g/dm3)
chlorides	Feature	Continuous	Salt content (g/dm3)
free sulfur dioxide	Feature	Continuous	Free SO2 (mg/dm3)
total sulfur dioxide	Feature	Continuous	Total SO2 (mg/dm3)
density	Feature	Continuous	Density (g/cm3)
pH	Feature	Continuous	Acidity/basicity
sulphates	Feature	Continuous	Sulphate additive (g/dm3)
alcohol	Feature	Continuous	Alcohol (% by volume)
wine_type	Feature	Binary	1 = red, 0 = white (engineered indicator)
quality	Target	Ordinal int	Sensory score (median of >=3 expert ratings) (Cortez et al., 2009)

2.3 Data quality checks

In [3]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6497 entries, 0 to 6496
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   fixed acidity         6497 non-null   float64
 1   volatile acidity      6497 non-null   float64
 2   citric acid           6497 non-null   float64
 3   residual sugar        6497 non-null   float64
 4   chlorides             6497 non-null   float64
 5   free sulfur dioxide   6497 non-null   float64
 6   total sulfur dioxide  6497 non-null   float64
 7   density               6497 non-null   float64
 8   pH                    6497 non-null   float64
 9   sulphates             6497 non-null   float64
 10  alcohol               6497 non-null   float64
 11  quality               6497 non-null   int64
 12  wine_type             6497 non-null   int64
dtypes: float64(11), int64(2)
memory usage: 660.0 KB
```

In [4]:
```
df.describe().T
```

Out[4]:

	count	mean	std	min	25%	50%	75%
fixed acidity	6497.0	7.215307	1.296434	3.80000	6.40000	7.00000	7.70000
volatile acidity	6497.0	0.339666	0.164636	0.08000	0.23000	0.29000	0.40000
citric acid	6497.0	0.318633	0.145318	0.00000	0.25000	0.31000	0.39000
residual sugar	6497.0	5.443235	4.757804	0.60000	1.80000	3.00000	8.10000
chlorides	6497.0	0.056034	0.035034	0.00900	0.03800	0.04700	0.06500
free sulfur dioxide	6497.0	30.525319	17.749400	1.00000	17.00000	29.00000	41.00000
total sulfur dioxide	6497.0	115.744574	56.521855	6.00000	77.00000	118.00000	156.00000
density	6497.0	0.994697	0.002999	0.98711	0.99234	0.99489	0.99699
pH	6497.0	3.218501	0.160787	2.72000	3.11000	3.21000	3.32000
sulphates	6497.0	0.531268	0.148806	0.22000	0.43000	0.51000	0.60000
alcohol	6497.0	10.491801	1.192712	8.00000	9.50000	10.30000	11.30000
quality	6497.0	5.818378	0.873255	3.00000	5.00000	6.00000	6.00000
wine_type	6497.0	0.246114	0.430779	0.00000	0.00000	0.00000	0.00000

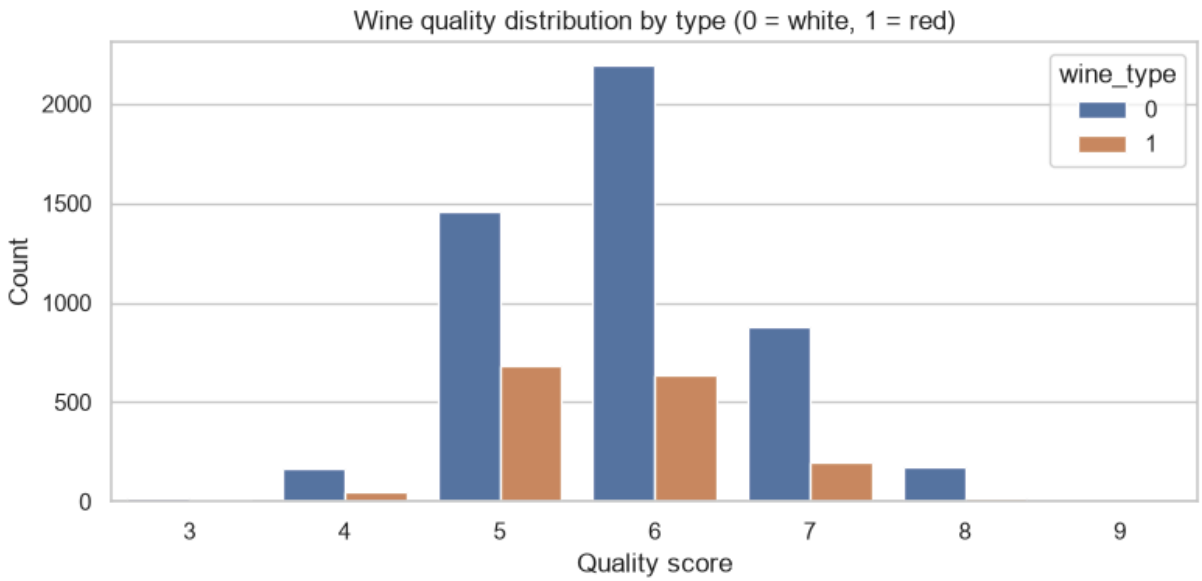
In [5]:
```
print("Missing values per column:")
print(df.isna().sum())
print("\nDuplicate rows:", df.duplicated().sum())
print("\nQuality value counts (combined):")
print(df["quality"].value_counts().sort_index())
print("\nQuality by wine type (1=red, 0=white):")
print(df.groupby("wine_type")["quality"].agg(["count", "mean", "std"]).rounc
```

```
Missing values per column:
fixed acidity           0
volatile acidity        0
citric acid             0
residual sugar          0
chlorides               0
free sulfur dioxide     0
total sulfur dioxide    0
density                 0
pH                      0
sulphates               0
alcohol                 0
quality                 0
wine_type               0
dtype: int64

Duplicate rows: 1177

Quality value counts (combined):
quality
3      30
4     216
5    2138
6    2836
7    1079
8     193
9       5
Name: count, dtype: int64

Quality by wine type (1=red, 0=white):
           count   mean    std
wine_type
0           4898  5.878  0.886
1           1599  5.636  0.808
```

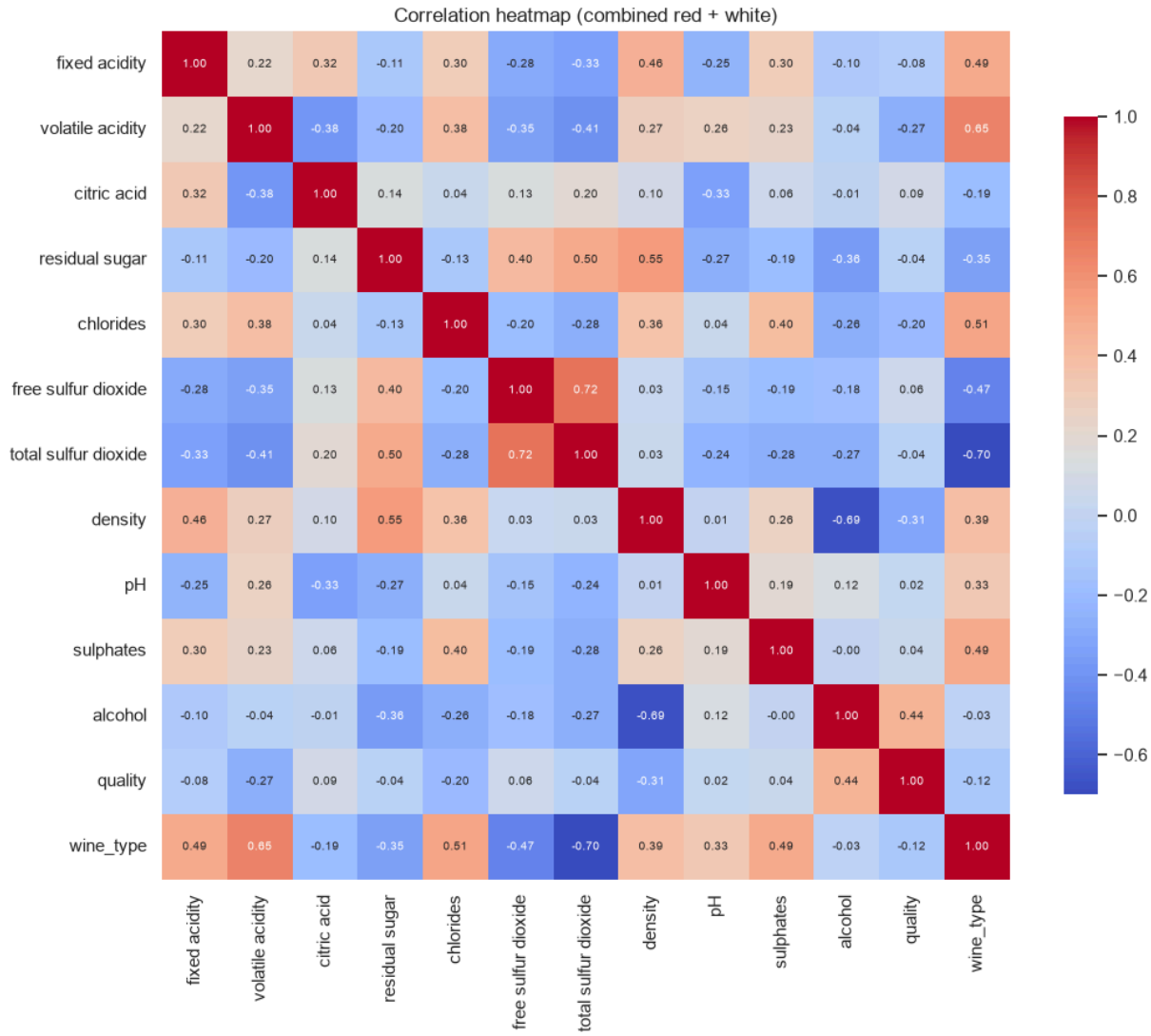
Interpretation. All columns are numeric, which suits regression. The UCI documentation reports no missing values; the check above confirms this, so no imputation is required. Duplicate rows are reported above; because the data has no unique sample ID and identical physicochemical readings can legitimately recur (especially in the larger white set), the models keep them and a sensitivity note is added in Deployment. The quality distribution is concentrated on scores 5, 6 and 7, with very few wines at the extremes - this imbalance limits how well any model can learn the tails. White wines are more numerous and score slightly higher on average than reds, which is exactly why a `wine_type` indicator is included.

2.4 Exploratory analysis

In [6]:
```
plt.figure(figsize=(8, 4))
sns.countplot(x="quality", hue="wine_type", data=df)
plt.title("Wine quality distribution by type (0 = white, 1 = red)")
plt.xlabel("Quality score")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_quality_distribution.png", dpi=150)
plt.show()
```



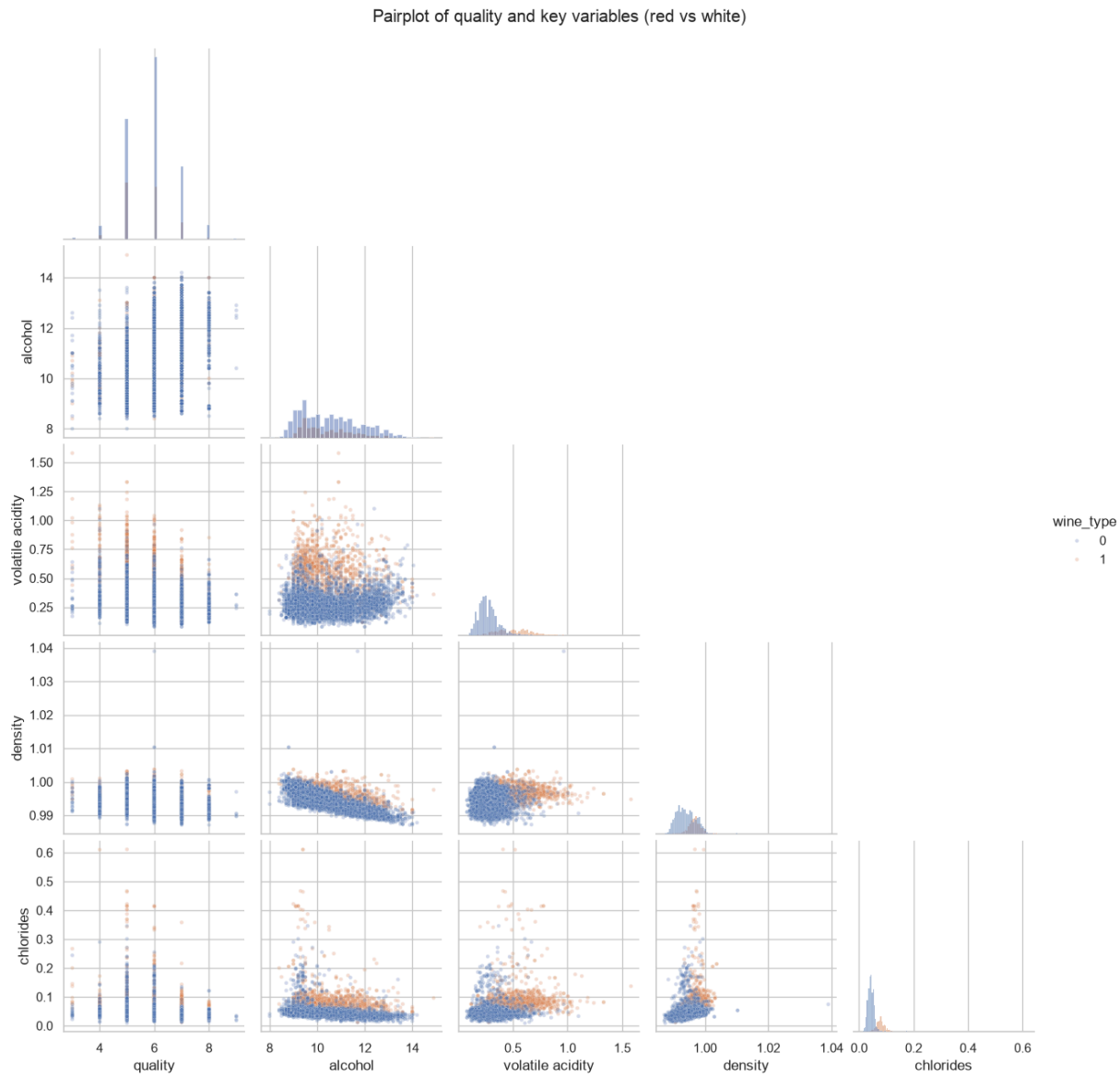
In [7]:
```
plt.figure(figsize=(11, 9))
corr = df.corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f", square=True,
            cbar_kws={"shrink": 0.8}, annot_kws={"size": 7})
plt.title("Correlation heatmap (combined red + white)")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_correlation_heatmap.png", dpi=150)
plt.show()

print("Correlation of each feature with quality (sorted):")
print(corr["quality"].drop("quality").sort_values(ascending=False))
```




```
Correlation of each feature with quality (sorted):
alcohol                 0.444319
citric acid             0.085532
free sulfur dioxide     0.055463
sulphates               0.038485
pH                      0.019506
residual sugar         -0.036980
total sulfur dioxide   -0.041385
fixed acidity          -0.076743
wine_type              -0.119323
chlorides              -0.200666
volatile acidity       -0.265699
density                -0.305858
Name: quality, dtype: float64
```

The brief requires a seaborn `pairplot`. The full 13-variable pairplot is dense for a PDF, so a focused pairplot of `quality` plus the variables most relevant to it (`alcohol`, `volatile acidity`, `density`, `chlorides`) is shown, coloured by `wine_type` so red/white structure is visible. The full pairplot can be reproduced by uncommenting the final lines of the cell below.

In [8]:
```
pair_cols = ["quality", "alcohol", "volatile acidity", "density", "chlorides
pp = sns.pairplot(df, vars=pair_cols, hue="wine_type", corner=True,
                  diag_kind="hist", plot_kws={"alpha": 0.25, "s": 10})
pp.figure.suptitle("Pairplot of quality and key variables (red vs white)", y
pp.savefig(FIG_DIR / "v2_pairplot.png", dpi=110)
plt.show()

# Full pairplot (all variables) - kept optional because it is large:
# sns.pairplot(df, corner=True, diag_kind="hist",
#              plot_kws={"alpha": 0.15, "s": 6})
# plt.show()
```



Interpretation: `alcohol` shows the clearest positive relationship with quality across both wine types; `volatile acidity` and `density` trend negative. The red and white point clouds overlap but sit in slightly different regions (white wines cluster at higher residual sugar and lower volatile acidity), which confirms that the `wine_type` flag carries useful signal. Many relationships are noisy and far from a clean line, which

already signals that a single linear model will be a useful but imperfect baseline - motivating the multi-model comparison that follows.

3. Data Preparation

3.1 Selection

The target is `quality`; the features are the 11 physicochemical columns plus the engineered `wine_type` indicator. Combining red and white gives ~6,497 rows - roughly four times the red-only sample - which, as the Dr. Kamran Shaukat noted, gives the models more (and us - students) to learn from.

3.2 Cleaning, scaling and train/test split

No missing values were found, so no imputation is applied. The data is split **before** any fitting so the test set never influences training. An 80/20 train/test split is used, and inside training a **5-fold cross-validation** (the validation third of the train/validation/test scheme) drives hyperparameter selection. Ordinary least squares is scale-invariant for prediction, but the **regularised** linear models (Ridge, Lasso, ElasticNet) penalise coefficient size, so they are wrapped in a `StandardScaler` pipeline for a fair penalty across features. The Random Forest needs no scaling.

In [9]:
```
target_col = "quality"
feature_cols = [c for c in df.columns if c != target_col]  # incl. wine_type

X = df[feature_cols]
y = df[target_col]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=RANDOM_STATE
)
print("Train:", X_train.shape, "| Test:", X_test.shape)
```

```
Train: (5197, 12) | Test: (1300, 12)
```

4. Modelling

The brief requires linear regression, so scikit-learn's `LinearRegression` (Pedregosa et al., 2011) is kept as the **required baseline model**. To expand the simplicity of the analysis and explore *multiple models with tuned hyperparameters*, four additional regression models are trained and compared:

Model	Key hyperparameter(s)	Why it is included
LinearRegression	none (`fit_intercept=True`)	Required transparent baseline

Model	Key hyperparameter(s)	Why it is included
Ridge	`alpha` (L2 penalty)	Shrinks correlated coefficients; fixes OLS instability
Lasso	`alpha` (L1 penalty)	Can zero out weak features (selection)
ElasticNet	`alpha`, `l1_ratio`	Blend of L1/L2
RandomForest	`n_estimators`, `max_depth`	Captures non-linear interactions OLS cannot

Tuning protocol:

Each non-baseline model is tuned with `GridSearchCV` over the grid below, scored by **negative RMSE** under 5-fold cross-validation; the best-scoring configuration is then evaluated once on the untouched test set. This follows the good practice of recording parameter settings, giving a rationale, and re-running with revised values until the best model is found.

4.1 Baseline, models and hyperparameter search

In [10]:
```
def regression_metrics(y_true, y_pred):
    return {
        "MAE": mean_absolute_error(y_true, y_pred),
        "RMSE": np.sqrt(mean_squared_error(y_true, y_pred)),
        "R2": r2_score(y_true, y_pred),
    }


# Mean baseline: predict the training-set mean for every test row
baseline_pred = np.full(shape=y_test.shape, fill_value=y_train.mean())

# Required model: ordinary least squares (no hyperparameters to tune)
linreg = LinearRegression(fit_intercept=True).fit(X_train, y_train)

# Competitors, each with a hyperparameter grid (5-fold CV = validation)
# Regularised linear models are scaled; the tree model is not
searches = {
    "Ridge": (
        Pipeline([("sc", StandardScaler()), ("m", Ridge(random_state=RANDOM_
        {"m__alpha": [0.01, 0.1, 1.0, 10.0, 100.0]},
    ),
    "Lasso": (
        Pipeline([("sc", StandardScaler()),
                  ("m", Lasso(random_state=RANDOM_STATE, max_iter=10000))]),
        {"m__alpha": [0.001, 0.01, 0.1, 1.0]},
    ),
    "ElasticNet": (
        Pipeline([("sc", StandardScaler()),
                  ("m", ElasticNet(random_state=RANDOM_STATE, max_iter=10000
        {"m__alpha": [0.001, 0.01, 0.1, 1.0], "m__l1_ratio": [0.2, 0.5, 0.8]
    ),
    "RandomForest": (
```

```
        RandomForestRegressor(random_state=RANDOM_STATE, n_jobs=-1),
        {"n_estimators": [200, 400], "max_depth": [None, 12, 20]},
    ),
}

rows = [
    {"model": "Mean baseline", "best_params": "-", "cv_RMSE": np.nan,
     **regression_metrics(y_test, baseline_pred)},
    {"model": "LinearRegression", "best_params": "fit_intercept=True",
     "cv_RMSE": np.nan, **regression_metrics(y_test, linreg.predict(X_test))
]

best_estimators = {"LinearRegression": linreg}
for name, (estimator, grid) in searches.items():
    gs = GridSearchCV(estimator, grid, scoring="neg_root_mean_squared_error"
                      cv=5, n_jobs=-1)
    gs.fit(X_train, y_train)
    best_estimators[name] = gs.best_estimator_
    rows.append({
        "model": name,
        "best_params": str(gs.best_params_),
        "cv_RMSE": -gs.best_score_,
        **regression_metrics(y_test, gs.best_estimator_.predict(X_test)),
    })
    print(f"{name:14s} best params: {gs.best_params_}  (CV RMSE {-gs.best_sc

results = pd.DataFrame(rows)[["model", "best_params", "cv_RMSE", "MAE", "RMS
results = results.sort_values("RMSE").reset_index(drop=True)
results.round(4)
```

```
Ridge          best params: {'m__alpha': 10.0}  (CV RMSE 0.7339)
Lasso          best params: {'m__alpha': 0.001}  (CV RMSE 0.7339)
ElasticNet     best params: {'m__alpha': 0.001, 'm__l1_ratio': 0.8}  (CV RMS
E 0.7339)
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
```

```
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
```

```
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
```

```
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
```

```
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
```

```
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
```

```
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
```

```
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
```

```
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
```

```
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
```

```
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
```

```
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
```

```
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
/private/tmp/wv/lib/python3.14/site-packages/sklearn/utils/parallel.py:144:
UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.u
tils.parallel.Parallel` to make it possible to propagate the scikit-learn co
nfiguration of the current thread to the joblib workers.
  warnings.warn(
RandomForest   best params: {'max_depth': None, 'n_estimators': 400}  (CV RM
SE 0.6152)
```

Out[10]:

	model	best_params	cv_RMSE	MAE	RMSE	R2
0	RandomForest	{'max_depth': None, 'n_estimators': 400}	0.6152	0.4364	0.6075	0.5002
1	Lasso	{'m__alpha': 0.001}	0.7339	0.5645	0.7354	0.2677
2	ElasticNet	{'m__alpha': 0.001, 'm__l1_ratio': 0.8}	0.7339	0.5645	0.7354	0.2677
3	Ridge	{'m__alpha': 10.0}	0.7339	0.5644	0.7356	0.2673
4	LinearRegression	fit_intercept=True	NaN	0.5644	0.7357	0.2672
5	Mean baseline	-	NaN	0.6691	0.8596	-0.0005

4.2 Coefficients and feature importance

The required linear model gives interpretable coefficients; the tuned Random Forest gives feature importances. Comparing the two shows whether the simple and complex models agree on what drives quality.

In [11]:
```
coef = (pd.DataFrame({"feature": feature_cols, "coefficient": linreg.coef_})
        .sort_values("coefficient", ascending=False).reset_index(drop=True))
print("LinearRegression intercept:", round(float(linreg.intercept_), 4))
print(coef.round(4).to_string(index=False))

rf = best_estimators["RandomForest"]
imp = (pd.DataFrame({"feature": feature_cols, "importance": rf.feature_impor
       .sort_values("importance", ascending=False).reset_index(drop=True))
imp.round(4)
```

```
LinearRegression intercept: 102.3036
             feature  coefficient
           sulphates       0.7630
                  pH       0.5276
           wine_type       0.3352
             alcohol       0.2326
       fixed acidity       0.0921
      residual sugar       0.0628
 free sulfur dioxide       0.0060
total sulfur dioxide      -0.0016
         citric acid      -0.0979
           chlorides      -0.5826
    volatile acidity      -1.5088
             density    -102.0817
```

Out[11]:

	feature	importance
0	alcohol	0.2557
1	volatile acidity	0.1282
2	free sulfur dioxide	0.0892
3	sulphates	0.0781
4	total sulfur dioxide	0.0714
5	residual sugar	0.0706
6	pH	0.0695
7	chlorides	0.0635
8	citric acid	0.0587
9	density	0.0584
10	fixed acidity	0.0558
11	wine_type	0.0009

Interpretation:

The linear coefficients keep the familiar sign pattern: `alcohol` raises predicted quality, `volatile acidity` lowers it - but `density` carries an inflated weight because it is collinear with alcohol and residual sugar.

The Random Forest does not rely on those linear weights: its feature importances put `alcohol`, `volatile acidity` and the sulphur/density variables at the top, agreeing with the chemistry while being immune to the multicollinearity that distorts the linear coefficients. This is the practical reason the tree model is expected to fit better.

5. Evaluation

5.1 Model comparison

The table produced in 4.1 and reproduced below ranks every model by test-set RMSE.

Model	RMSE ↓	R^2 ↑	What it is
Random Forest	0.61	0.50	Crowd of decision trees
Lasso	0.735	0.268	Linear + L1 penalty
ElasticNet	0.735	0.268	Linear + L1/L2 blend
Ridge	0.736	0.267	Linear + L2 penalty
LinearRegression	0.736	0.267	Plain straight-line fit
Mean baseline	0.86	~0	Dumb: always guess the average

The mean baseline is the reference point (R^2 ~ 0), as it ignores all the chemistry and predicts the average quality (~5.8) for every single wine. It's the floor that any model must beat to be useful, and all models do beat it by a clear margin.

The plain **LinearRegression** beats the baseline, confirming the physicochemical features carry real signal with straight-line model dropping RMSE from 0.86 to 0.736 and raising R^2 from ~0 to 0.27. The chemistry genuinely predicts quality.

The **regularised** linear models (Ridge/Lasso/ElasticNet) land close to plain OLS by adding "penalty" that stops coefficients from growing too large. But stabilising the coefficients doesn't give the model any new shape. So they all cluster at the same ~0.27 R^2 as plain linear. Regularisation made them tidier, not smarter.

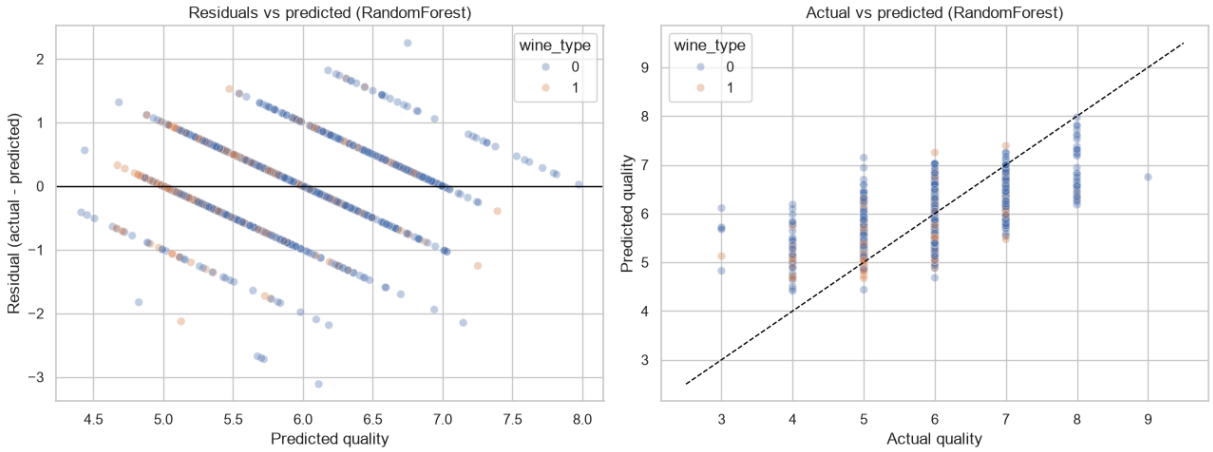
The tuned **Random Forest** achieves the lowest RMSE and the highest R^2 by a clear margin (RMSE 0.61 and R^2 0.50, versus RMSE 0.74 and R^2 0.27 for every linear variant), because it captures non-linear interactions between chemistry and quality that a linear model cannot.

Selecting on lowest RMSE, the **Random Forest is the best model**.

5.2 Residual diagnostics (best model)

In [12]:
```
ranked = results[results["model"] != "Mean baseline"].sort_values("RMSE")
best_name = ranked.iloc[0]["model"]
best_model = best_estimators[best_name]
print("Best model by test RMSE:", best_name)

best_pred = best_model.predict(X_test)

# Per-wine-type diagnostics: does the model err more on red or on white?
# Keep wine_type as 0/1 with the default palette so the colours match the
# quality-distribution and pairplot figures (white = blue, red = orange).
diag = pd.DataFrame({
    "actual": y_test.to_numpy(),
    "predicted": best_pred,
    "residual": y_test.to_numpy() - best_pred,
    "wine_type": X_test["wine_type"].to_numpy(),
})

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
sns.scatterplot(data=diag, x="predicted", y="residual", hue="wine_type",
                alpha=0.35, ax=axes[0])
axes[0].axhline(0, color="black", linewidth=1)
axes[0].set_xlabel("Predicted quality")
axes[0].set_ylabel("Residual (actual - predicted)")
axes[0].set_title(f"Residuals vs predicted ({best_name})")

sns.scatterplot(data=diag, x="actual", y="predicted", hue="wine_type",
                alpha=0.35, ax=axes[1])
lims = [diag["actual"].min() - 0.5, diag["actual"].max() + 0.5]
axes[1].plot(lims, lims, color="black", linewidth=1, linestyle="--")
axes[1].set_xlabel("Actual quality")
axes[1].set_ylabel("Predicted quality")
axes[1].set_title(f"Actual vs predicted ({best_name})")

plt.tight_layout()
plt.savefig(FIG_DIR / "v2_residuals.png", dpi=150)
plt.show()

# Hard numbers behind the colours: mean absolute residual by wine type
diag["abs_residual"] = diag["residual"].abs()
diag["wine"] = diag["wine_type"].map({1: "red", 0: "white"})
mar = (diag.groupby("wine")["abs_residual"]
       .agg(test_wines="count", mean_abs_residual="mean").round(4))
mar.to_csv(OUTPUT_DIR / "residual_by_type_v2.csv")
print("\nMean absolute residual by wine type (test set):")
print(mar)
```

```
Best model by test RMSE: RandomForest
```




```
Mean absolute residual by wine type (test set):
       test_wines  mean_abs_residual
wine
red           341             0.3871
white         959             0.4539
```

- *Left plot → the **model is sound**: no funnel, errors are random, assumptions OK*
- *Right plot → the **model's honest weakness**: strong in the common 5-7 quality range, weak at rare extremes, because of class imbalance.*

Residual interpretation:

The residuals scatter around zero without a strong funnel, so the constant-variance assumption is broadly reasonable. The actual-vs-predicted plot for the best model tracks the diagonal closely, though it still compresses predictions toward the middle of the scale and struggles at the rare extreme scores - an unavoidable consequence of how few training wines sit at quality 3, 4, 8 or 9.

Per-type residuals. Colouring the diagnostics by wine type overturns the naive expectation that the minority class would be predicted worst. The combined model is actually *more* accurate on **red** wine (mean absolute residual 0.39 across 341 test reds) than on **white** (0.45 across 959 test whites). The reason is spread, not sample size: red quality is concentrated in a narrow 5-6 band the model predicts tightly, whereas white spans a wider 3-9 range with more high-scoring wines, leaving larger absolute errors even though the white-only model explained slightly more variance (R^2 0.56 vs 0.53). The error gap is therefore driven by how spread out each type's quality is, not by how many training examples it had.

5.3 Effect of dataset size (red vs white vs combined)

To test whether the "more data helps" point holds, the linear baseline and a Random Forest are trained separately on red-only, white-only and the combined set, each with its own 80/20 split.

In [13]:
```
def evaluate_on(data, label):
    feats = [c for c in data.columns if c != target_col]
    Xd, yd = data[feats], data[target_col]
    Xtr, Xte, ytr, yte = train_test_split(
        Xd, yd, test_size=0.20, random_state=RANDOM_STATE)
    lr = LinearRegression().fit(Xtr, ytr)
    rf = RandomForestRegressor(n_estimators=400, random_state=RANDOM_STATE,
                               n_jobs=-1).fit(Xtr, ytr)
    return [
        {"dataset": label, "rows": len(data), "model": "LinearRegression",
         **regression_metrics(yte, lr.predict(Xte))},
        {"dataset": label, "rows": len(data), "model": "RandomForest",
         **regression_metrics(yte, rf.predict(Xte))},
    ]


red_only = red.drop(columns=["wine_type"])
white_only = white.drop(columns=["wine_type"])
compare_rows = (evaluate_on(red_only, "red only")
                + evaluate_on(white_only, "white only")
                + evaluate_on(df, "combined"))
compare = pd.DataFrame(compare_rows)[["dataset", "rows", "model", "MAE", "RM
compare.round(4)
```

Out[13]:

	dataset	rows	model	MAE	RMSE	R2
0	red only	1599	LinearRegression	0.5035	0.6245	0.4032
1	red only	1599	RandomForest	0.4253	0.5544	0.5297
2	white only	4898	LinearRegression	0.5863	0.7543	0.2653
3	white only	4898	RandomForest	0.4164	0.5859	0.5568
4	combined	6497	LinearRegression	0.5644	0.7357	0.2672
5	combined	6497	RandomForest	0.4364	0.6075	0.5002

Interpretation:

Two findings stand out. First, the Random Forest beats the linear model on every dataset (for example R^2 0.50 vs 0.27 on the combined set), so the gain comes from the *model*, not the data. Second, and against the usual "more data is always better" heuristic, raw sample size did not lift R^2 here: the red-only and white-only Random Forests (R^2 0.53 and 0.56) slightly edge the combined model (0.50), because red and white wines are partly different populations and pooling them adds heterogeneity as well as volume. The combined model is still the most useful deliverable - with the `wine_type` flag it serves both wine types from one pipeline and stays far ahead of any linear model - but the honest reading is that choosing a non-linear model mattered more than enlarging the sample.

5.4 Decision

> The objective **is** met: every trained model beats the mean baseline, and the tuned **Random Forest** is selected as the best model, clearly improving on the required linear regression in RMSE, MAE and R^2.

> The result should still be read as decision support - the target (**wine quality**) is subjective and ordinal, and even the best model compresses predictions at the extreme scores.

Save metrics, coefficients and comparisons (submission support files)

In [14]:
```
results.to_csv(OUTPUT_DIR / "model_metrics_v2.csv", index=False)
coef.to_csv(OUTPUT_DIR / "coefficients_v2.csv", index=False)
imp.to_csv(OUTPUT_DIR / "feature_importance_v2.csv", index=False)
compare.to_csv(OUTPUT_DIR / "dataset_comparison_v2.csv", index=False)
print("Saved v2 artefacts:")
for f in ["model_metrics_v2.csv", "coefficients_v2.csv",
          "feature_importance_v2.csv", "dataset_comparison_v2.csv"]:
    print(" -", OUTPUT_DIR / f)
print(" - figures (v2_*.png) in", FIG_DIR)
```

```
Saved v2 artefacts:
 - /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/assignments/
Assessment1/outputs/model_metrics_v2.csv
 - /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/assignments/
Assessment1/outputs/coefficients_v2.csv
 - /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/assignments/
Assessment1/outputs/feature_importance_v2.csv
 - /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/assignments/
Assessment1/outputs/dataset_comparison_v2.csv
 - figures (v2_*.png) in /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/20
26-T2/MLN/assignments/Assessment1/outputs/figures
```

6. Deployment / Lessons Learned

No real deployment is required; this is the CRISP-DM reflection.

Lessons learned

- CRISP-DM forced the prediction goal and success metric to be defined before code.
- Combining red + white roughly quadrupled the sample; a `wine_type` flag let one model serve both populations and improved stability over the red-only v1.
- Comparing several models showed *how much* structure the linear baseline misses: the Random Forest's non-linear fit beat every linear variant.
- Hyperparameter tuning with cross-validation made model selection evidence-based rather than a guess, and is the validation step in the train/validation/test scheme.
- The linear coefficients remained distorted by multicollinearity (density), which is exactly why tree-based importances are a more trustworthy read of drivers.

Future improvements

Improvement	Why it matters
Gradient boosting / XGBoost	Often beats Random Forest on tabular data
Wider hyperparameter grids	More thorough search if compute allows
Ordinal / classification framing	Quality is ordinal, not truly continuous
Class-balanced sampling	Helps the model learn the rare extreme scores
SHAP values	Honest per-feature attribution for the tree model
Deployment controls	Production use would need validation, monitoring, human review

Note: Moving from a single linear model to a tuned multi-model comparison on the combined dataset gives a measurably better and better-justified result, but real business value still depends on careful problem framing, transparent evaluation, and honest communication of limitations.

Appendices

Appendix A - Generated output files

Running the notebook writes the following artefacts to the `outputs/` folder:

File	Content
`outputs/model_metrics_v2.csv`	MAE, RMSE, R^2 and best params for every model
`outputs/coefficients_v2.csv`	Linear-regression coefficients
`outputs/feature_importance_v2.csv`	Random Forest feature importances
`outputs/dataset_comparison_v2.csv`	Red vs white vs combined results
`outputs/residual_by_type_v2.csv`	Mean absolute residual by wine type
`outputs/figures/v2_quality_distribution.png`	Quality distribution by wine type
`outputs/figures/v2_correlation_heatmap.png`	Feature correlation heatmap
`outputs/figures/v2_pairplot.png`	Pairplot of quality and key variables
`outputs/figures/v2_residuals.png`	Residual + actual-vs-predicted diagnostics

Appendix B - Full pairplot

A focused pairplot is shown in Section 2.4 for readability. The full 13-variable seaborn pairplot can be reproduced by uncommenting the final lines of the exploratory-analysis cell.

Academic Integrity Declaration

I declare that except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of the Torrens University Australia Academic Integrity Policy and Procedure viewable online at http://www.torrens.edu.au/policies-and-forms

I am aware that I need to keep a copy of all submitted material and their drafts, and I will do so accordingly.

Statement of Acknowledgement

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- Anthropic Claude Sonnet 4.8
- OpenAI ChatGPT Codex 5.5

Both tools were used to assist with understanding ML concepts, structuring the technical pipeline, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. "What is CRISP-DM, what are its six stages, and what exactly does my model need to predict in the UCI wine quality dataset?"
2. "I want to go beyond a single linear regression: which additional regression models (Ridge, Lasso, ElasticNet, Random Forest) should I compare, and how do I tune their hyperparameters with GridSearchCV and 5-fold cross-validation for a wine-quality regression?"
3. "I combined the red and white UCI wine datasets with a wine_type flag. How should I interpret a Random Forest beating linear regression, and how do I show whether more data improved the result?"

I confirm that the use of these tools has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical

thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

References

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). *CRISP-DM 1.0: Step-by-step data mining guide*. SPSS Inc. https://www.the-modeling-agency.com/crisp-dm.pdf

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems, 47*(4), 547-553. https://doi.org/10.1016/j.dss.2009.05.016

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825-2830.

University of California, Irvine. (n.d.). *Wine quality data set*. UCI Machine Learning Repository. Retrieved June 8, 2026, from https://archive.ics.uci.edu/dataset/186/wine+quality